# Cyrus Guided Notes: Tsinghua control spine

Tsinghua course pack

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## Course source

**Official source:** `https://www.au.tsinghua.edu.cn/`

**Why this course is in the system.** Connect automatic control, modern control, optimization, estimation, robust control, and nonlinear control.

**Learning output.** Generate individual guided PDFs in batches, with the roadmap as the current index.

## Prerequisite checkpoint

- Differential equations
- Signals and systems
- Matrix rank and eigenvalues

## Formula checkpoint

Do not memorize these first. Read each formula as a sentence: what changes, what is observed, what is optimized, and what output it produces.

$$\mathcal{C} = [B \; AB \; \cdots \; A^{n-1}B], \quad \operatorname{rank}(\mathcal{C}) = n$$

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

$$V(\boldsymbol{x}) = \boldsymbol{x}^{\mathsf{T}} P \boldsymbol{x}, \quad \dot{V}(\boldsymbol{x}) < 0$$

$$A^{\mathsf{T}} P + PA - PBR^{-1}B^{\mathsf{T}} P + Q = 0$$

## Visual page

Draw controllability, observability, stability, and LQR as four connected gates.

## GoodNotes pages

- Page THU-C001: controllability and observability
- Page THU-C002: Lyapunov and LQR

## Web interaction

A rank-gate visual where the learner toggles B and C and watches rank change.

## Obsidian and Notion

| Layer | Output |
| --- | --- |
| Obsidian | Control Engineering -> Tsinghua control spine |
| Notion | Mastery levels for control gates. |

## First study move

Open the official source, copy the prerequisite checkpoint into GoodNotes, then write one formula in your own words before watching or reading more.